PROJECT EVIDENCE / 2026

AI/ML Salary Prediction

A visual record of the dataset, exploratory analysis, preprocessing, regression experiments, evaluation and sample prediction.

SOURCE RECORDS	MODEL INPUTS	MODELS TESTED	TOP TEST R^2
500	11	4	0.935
AI/ML job listings	encoded features	regression models	Gradient Boosting

WORKFLOW RECORDED

Load data → Explore → Encode → Split → Train → Evaluate → Predict

IN THIS RECORD

01 Dataset preview and missing-value audit

02 Salary distribution, correlation, and categorical comparisons

03 Encoding and train/test split summary

04 Four regression models with MAE, RMSE and R^2

05 Best-model diagnostics, Random Forest feature importance, and a sample prediction

Data note: 166 blank cells in total; 166 are in Certification. The current notebook treats these as a categorical 'nan' value.

Dataset and data quality

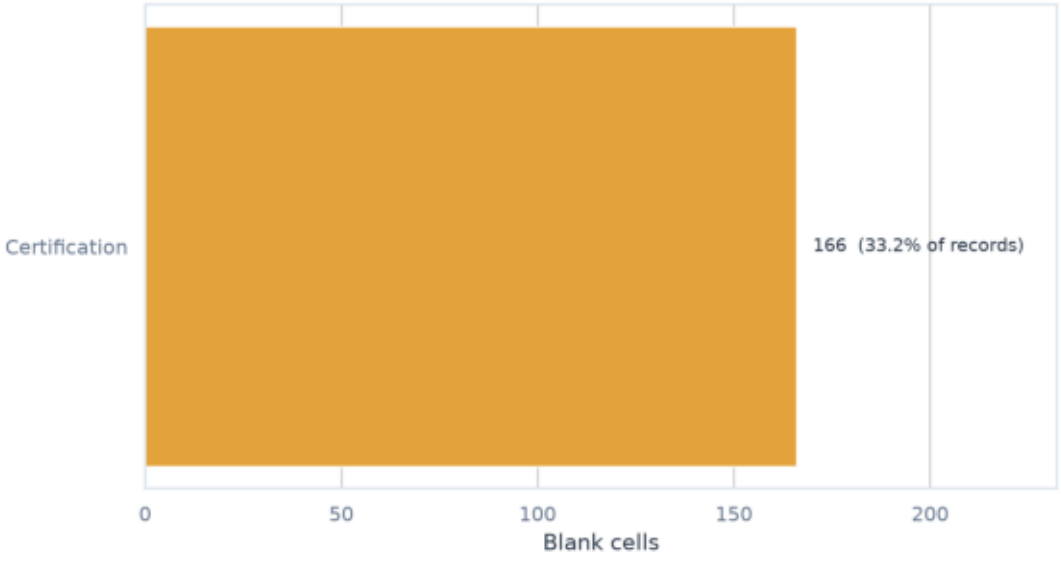
Source: ai_ml_job_analysis.csv · 500 rows · 11 columns

FIGURE 01 | SOURCE DATASET PREVIEW · FIRST 6 RECORDS

500 records • 11 source columns • Target: Salary (USD)

ID	Job Title	Experience (yrs)	Education	Industry	Location	Remote Type	Salary USD
1	Research Intern	2.2	PhD	Technology	Toronto, Canada	Yes	35800
2	Machine Learning Engineer	11.8	Bachelors	Finance	Mumbai, India	Hybrid	34400
3	AI Consultant	13.0	Masters	Technology	London, UK	No	143200
4	Applied Scientist	3.2	Bachelors	Automotive	Mumbai, India	No	24700
5	NLP Engineer	16.9	Masters	Healthcare	Seattle, USA	Yes	170200
6	AI Product Manager	5.9	Masters	Entertainment	Toronto, Canada	Yes	111900

FIGURE 02 | MISSING-VALUE AUDIT · 166 BLANK CELLS



Only fields with missing entries are plotted; every other source field has zero blanks.

Exploratory analysis

Salary range, numeric relationships, and experience patterns in the source records.

FIGURE 03 | SALARY DISTRIBUTION

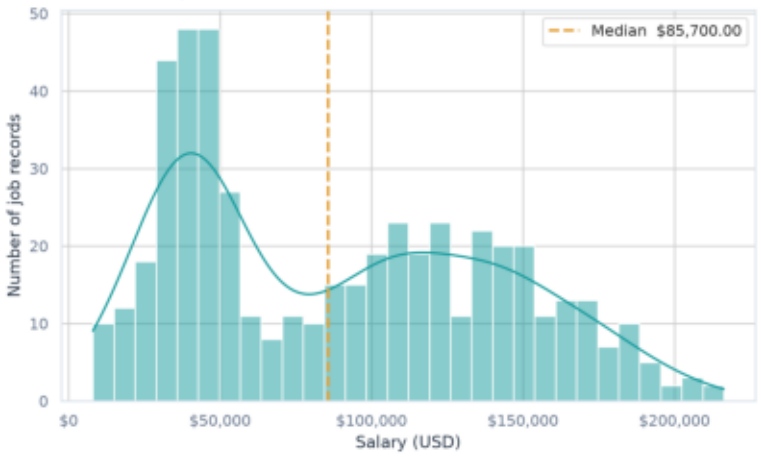


FIGURE 04 | NUMERIC-FEATURE CORRELATION HEATMAP

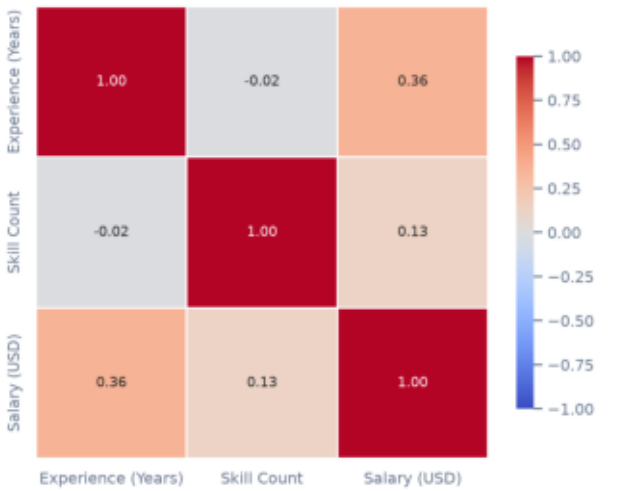
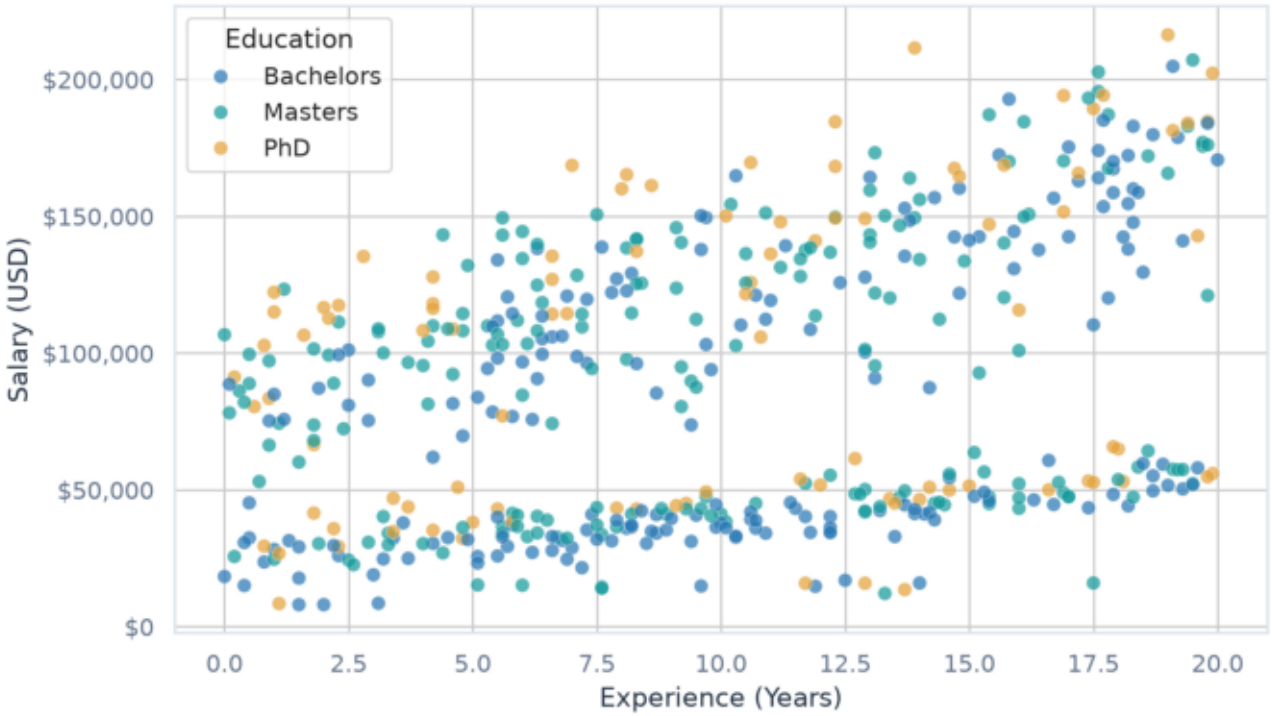


FIGURE 05 | EXPERIENCE VS. SALARY



Salary by work and education

Box plots compare the spread and median salary across two categorical variables.

FIGURE 06 | SALARY BY EDUCATION

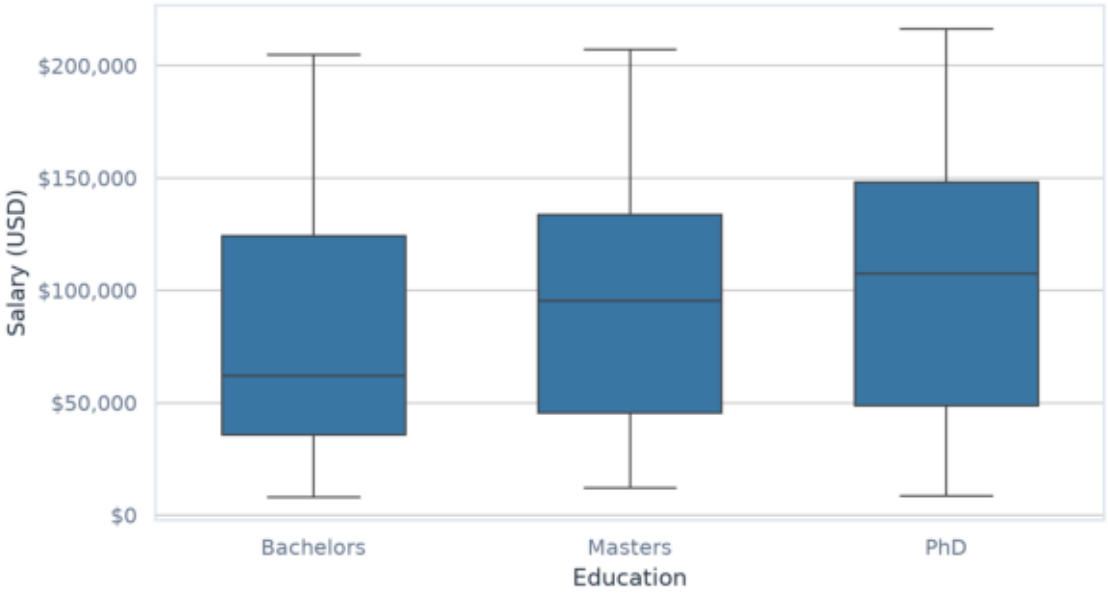
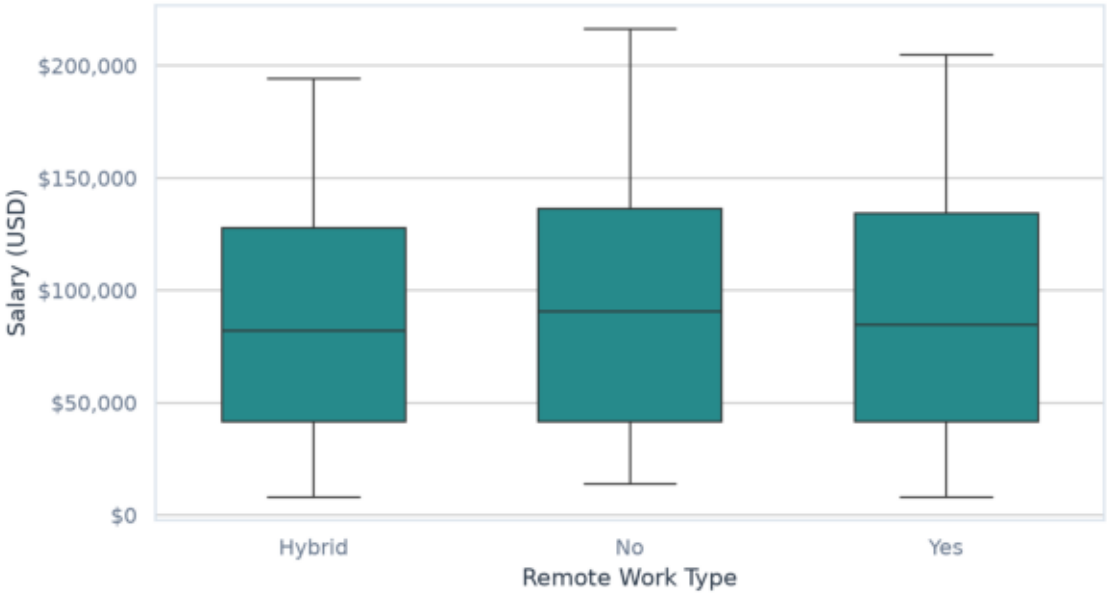


FIGURE 07 | SALARY BY REMOTE-WORK TYPE



Salary by role and industry

Job titles are ordered by frequency; industry categories are shown for comparison.

FIGURE 08 | SALARY BY JOB TITLE

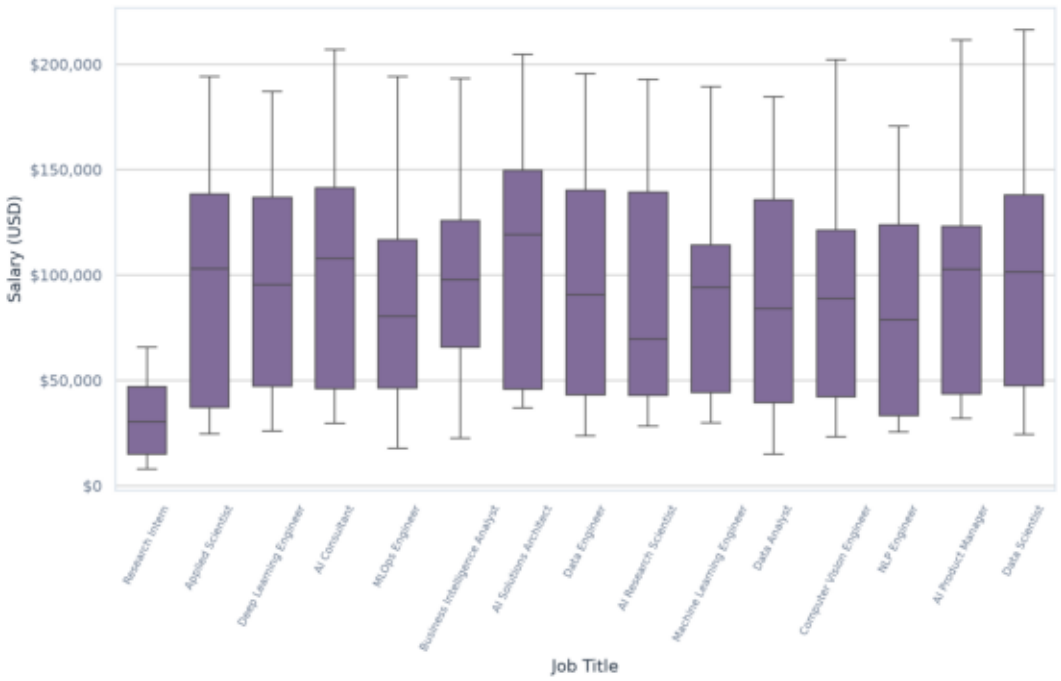
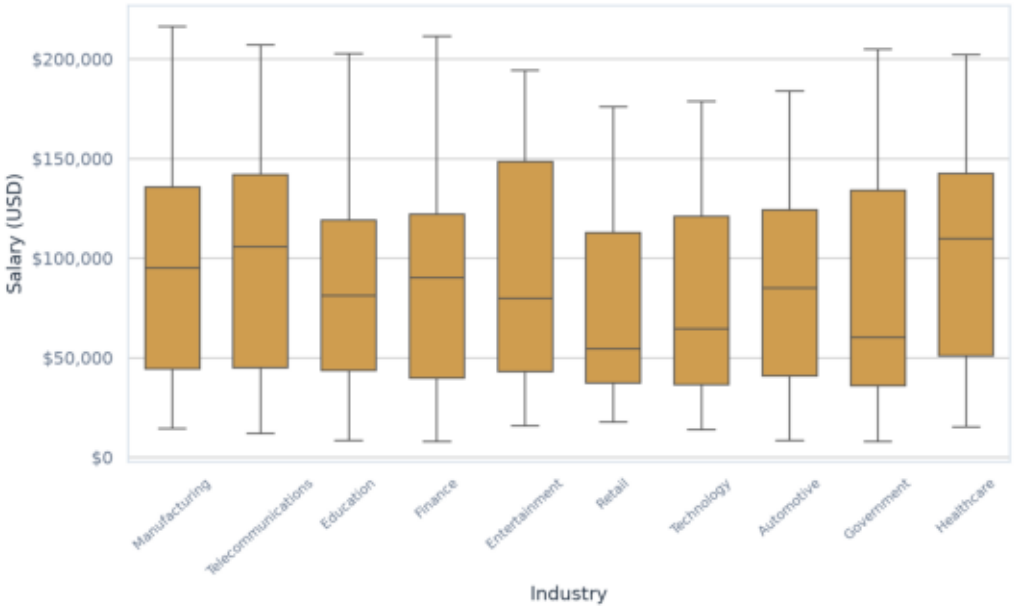


FIGURE 09 | SALARY BY INDUSTRY



Preprocessing and model outputs

The evaluation uses an 80/20 split with random_state = 42.

FIGURE 10 | PREPROCESSING AND FEATURE-MATRIX SUMMARY

ID	Dropped as a record identifier
Education	Ordinal map: Bachelors = 0 · Masters = 1 · PhD = 2
Remote Work	One-hot encoded into Remote_Hybrid, Remote_No, Remote_Yes
High-cardinality fields	Label encoded: Job Title, Industry, Company, Location, Certification
Train / test split	400 train · 100 test · random_state = 42
Model inputs	500 rows · 11 feature columns

Note: this reproduces the notebook's current categorical handling; blank certification values become a separate 'nan' label.

FIGURE 11 | REGRESSION MODEL METRICS ON THE HELD-OUT TEST SET

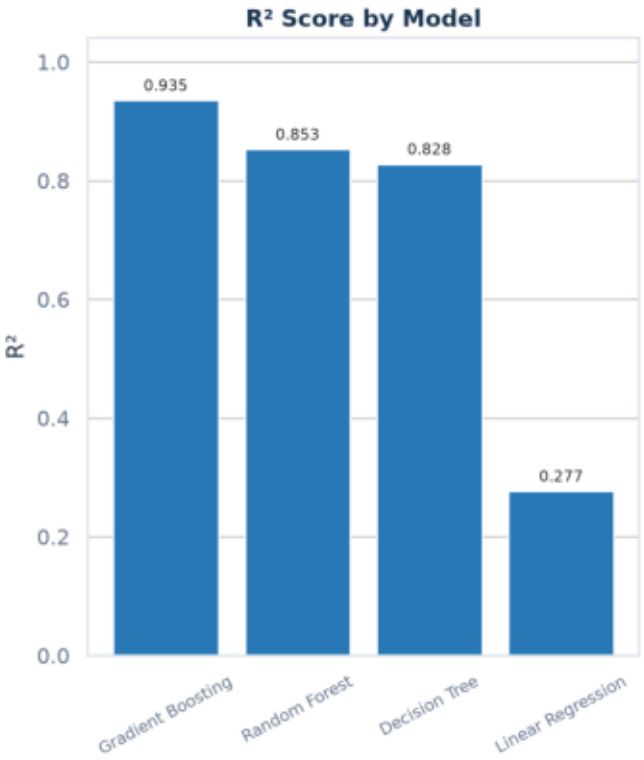
Model	MAE (USD) ↓	RMSE (USD) ↓	R^2 ↑
Gradient Boosting	$9,090.10	$12,646.38	0.9353
Random Forest	$13,886.34	$19,065.77	0.8529
Decision Tree	$14,211.95	$20,621.29	0.8280
Linear Regression	$36,748.75	$42,274.27	0.2770

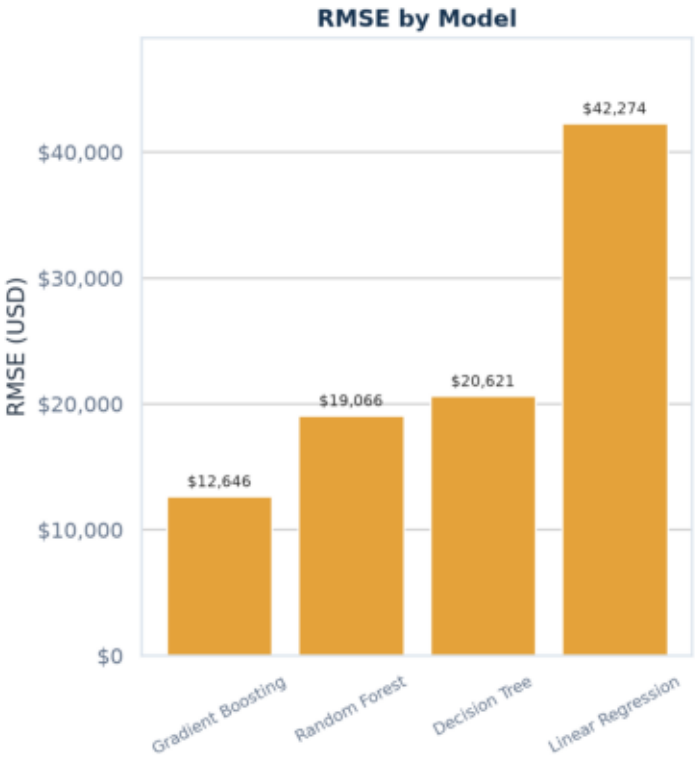
Lower MAE/RMSE and higher R^2 indicate better held-out fit; values are in USD except R^2.

Comparing four regressors

Decision Tree, Random Forest and Gradient Boosting improve on the linear baseline for this split.

FIGURE 12 | MODEL COMPARISON





BEST MODEL	TEST R²	MAE	RMSE
Gradient Boosting	**0.9353**	**$9,090.10**	**$12,646.38**

Best-model diagnostics

Predicted-vs-actual values and residuals for Gradient Boosting on held-out records.

FIGURE 13 | PREDICTED VS. ACTUAL SALARY · GRADIENT BOOSTING

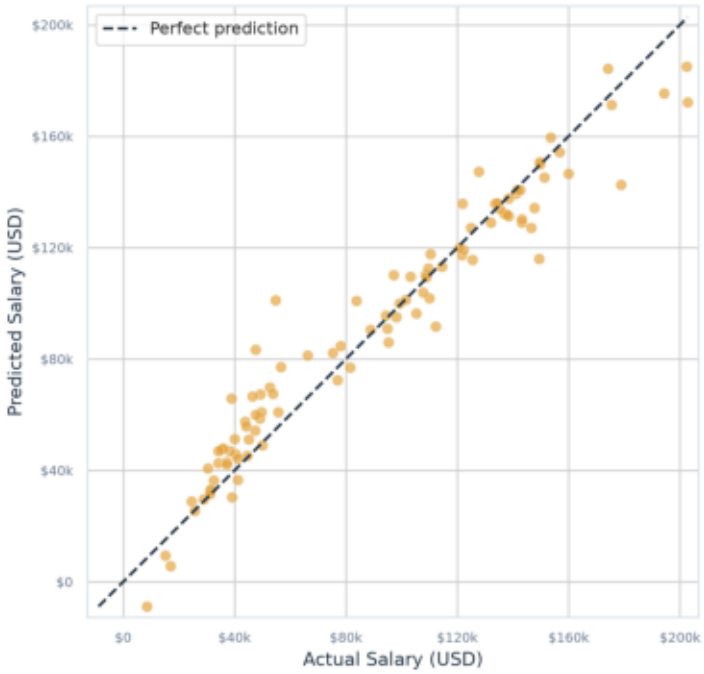
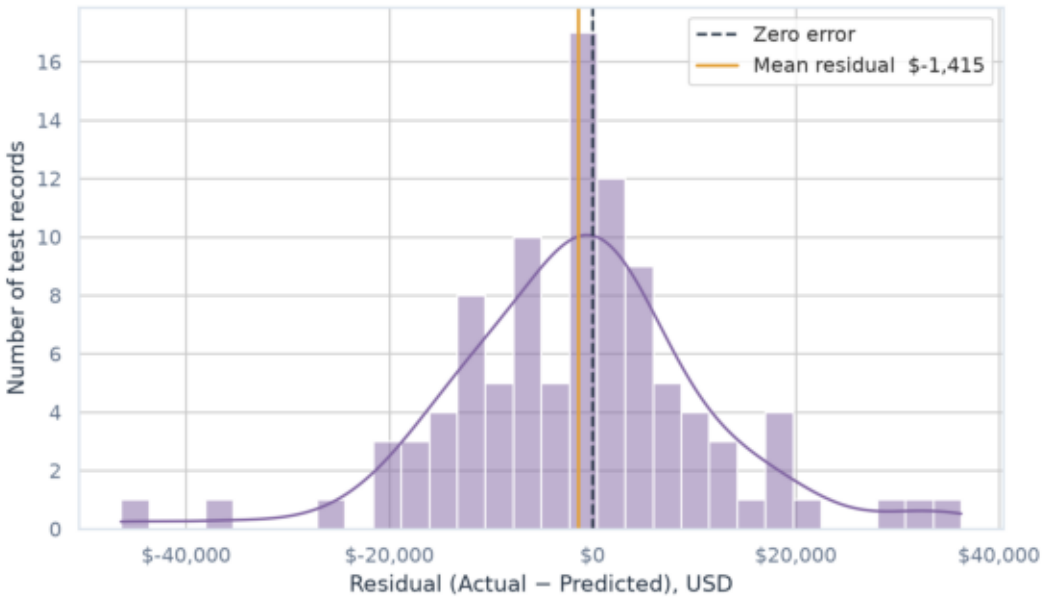


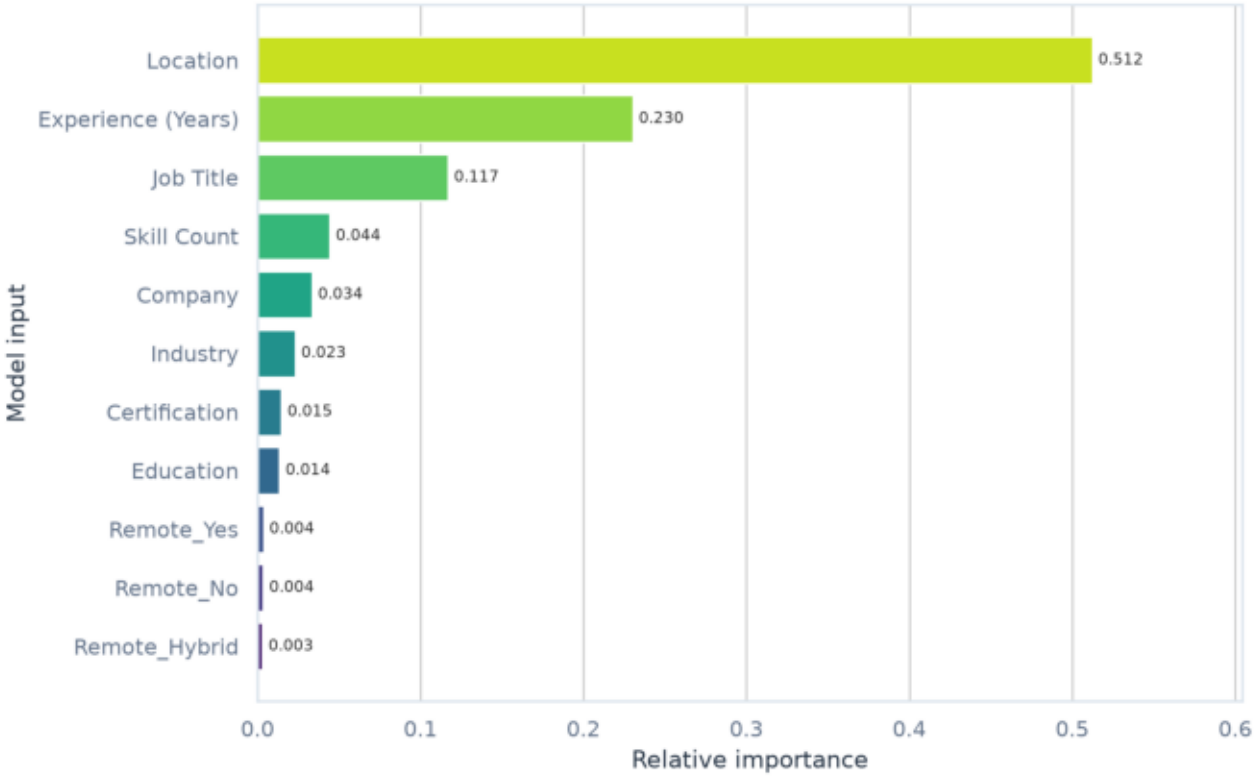
FIGURE 14 | RESIDUAL DISTRIBUTION · GRADIENT BOOSTING



Feature importance

Random Forest importances describe this fitted forest and are not causal effects.

FIGURE 15 | FEATURE IMPORTANCE · RANDOM FOREST



Importance ranks each encoded input by its contribution to the fitted Random Forest's split decisions. The notebook reports these values separately from the best-R^2 model.

Custom-profile prediction

A demonstration using the editable example profile from the notebook.

FIGURE 16 | SAMPLE CUSTOM-PROFILE SALARY PREDICTION

Random Forest estimate

$114,232.08

Job Title	Machine Learning Engineer	Industry / Company	Technology / Google
Experience	5.0 years	Location	Bengaluru, India
Education	Masters	Remote Work	Yes
Skills	12	Certification	AWS ML Specialty

Unseen categories use the notebook's first-known-label fallback: Location.

READING THIS OUTPUT

- The custom-profile cell uses the Random Forest model, while model comparison selects the best model by test-set R^2.
- In this example, the unseen Location category uses the notebook's first-known-label fallback.
- The result is a model estimate from this 500-row project dataset, not a salary guarantee or a production-ready valuation.
- Before deployment, explicitly impute missing certifications and use a training-safe categorical encoder with unknown-category handling.

Record complete: data loaded · EDA · encoding · split · 4 models · evaluation · diagnostics · feature importance · example prediction